

Администрирование сетевых подсистем

Лабораторная работа №5 — Конфигурирование HTTPS и PHP на Apache

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Цели и задачи работы

Цель лабораторной работы

Приобрести навыки конфигурирования веб-сервера Apache для работы по защищённому протоколу HTTPS и поддержки PHP.

Конфигурирование HTTPS

Создание SSL-ключа и сертификата

[illegible]

Настройка виртуального хоста Apache

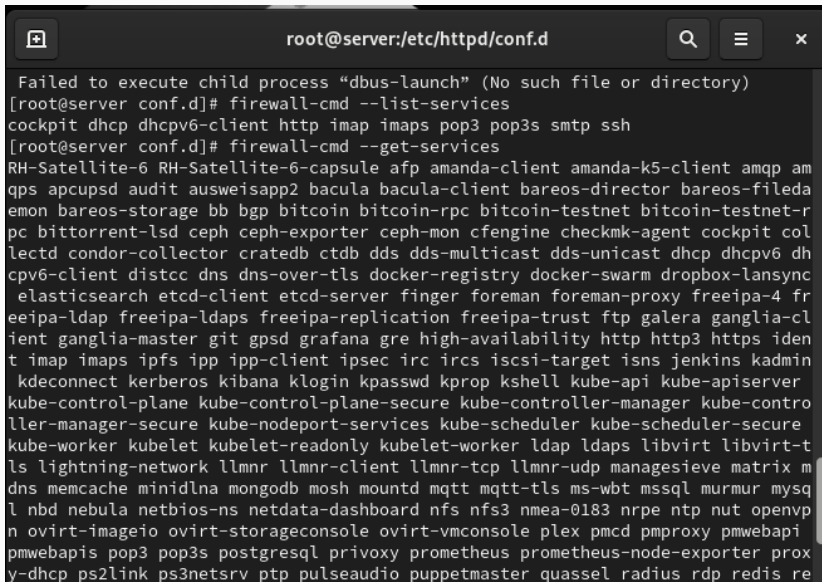
[illegible]

Перенаправление HTTP → HTTPS



Рис. 3: Перенаправление на HTTPS

Проверка работы HTTPS



A terminal window titled 'root@server:/etc/httpd/conf.d' with standard window controls (minimize, maximize, close). The terminal output shows an error message followed by two firewall commands and their results.

```
Failed to execute child process "dbus-launch" (No such file or directory)
[root@server conf.d]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client http imap imaps pop3 pop3s smtp ssh
[root@server conf.d]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp amqps
apcupsd audit ausweisapp2 bacula bacula-client bareos-director bareos-filedemon
bareos-storage bb bgp bitcoin bitcoin-rpc bitcoin-testnet bitcoin-testnet-rpc
bittorrent-lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent cockpit collectd
condor-collector cratedb ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client
distcc dns dns-over-tls docker-registry docker-swarm dropbox-lansync elasticsearch
etcd-client etcd-server finger foreman foreman-proxy freeipa-4 freeipa-ldap freeipa-ldaps
freeipa-replication freeipa-trust ftp galera ganglia-client ganglia-master git gpsd grafana
gre high-availability http http3 https ident imap imaps ipfs ipp ipp-client ipsec irc ircs
iscsi-target isns jenkins kadmin kdeconnect kerberos kibana klogin kpasswd kprop kshell
kube-api kube-apiserver kube-control-plane kube-control-plane-secure kube-controller-manager
kube-controller-manager-secure kube-nodeport-services kube-scheduler kube-scheduler-secure
kube-worker kubelet kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-tls
lightning-network llmnr llmnr-client llmnr-tcp llmnr-udp managesieve matrix memcached
memcache minidlna mongodb mosh mntd mqtt mqtt-tls ms-wbt mssql murmur mysql nbd nebula
netbios-ns netdata-dashboard nfs nfs3 nmap nmap-0183 nrpe ntp nut openvpn ovirt-imageio
ovirt-storageconsole ovirt-vmconsole plex pmcd pmproxy pmwebapi pmwebapis pop3 pop3s
postgres privoxy prometheus prometheus-node-exporter proxy-dhcp ps2link ps3netsrv ptp
pulseaudio puppetmaster quassel radius rdp redis re
```

Просмотр сертификата в браузере

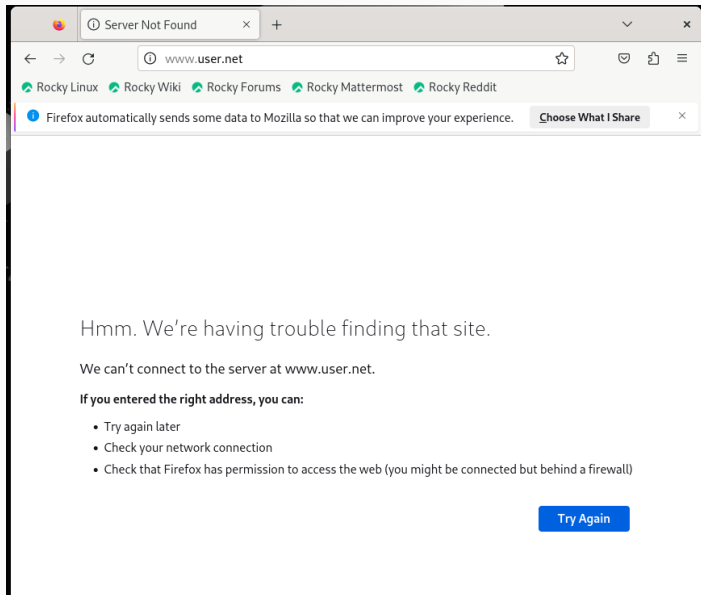
```
root@server:/etc/httpd/conf.d

dns memcache minidlna mongodb mosh mountd mqtt mqtt-tls ms-wbt mssql murmur mysq
l nbd nebula netbios-ns netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut openvp
n ovirt-imageio ovirt-storageconsole ovirt-vmconsole plex pmcd pmproxy pmwebapi
pmwebapis pop3 pop3s postgresql privoxy prometheus prometheus-node-exporter prox
y-dhcp ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radius rdp redis re
dis-sentinel rpc-bind rquotad rsh rsyncd rtsp salt-master samba samba-client sam
ba-dc sane sip sips slp smtp smtp-submission smtps snmp snmptls snmptls-trap smn
ptrap spideroak-lansync spotify-sync squid ssdp ssh steam-streaming svdrp svn sy
ncthing syncthing-gui syncthing-relay synergy syslog syslog-tls telnet tentacle
tftp tile38 tinc tor-socks transmission-client unpnp-client vdsm vnc-server warpi
nator wbem-http wbem-https wireguard ws-discovery ws-discovery-client ws-discove
ry-tcp ws-discovery-udp wsman wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp
-server zabbix-agent zabbix-server zerotier

[root@server conf.d]# firewall-cmd --add-service=https
success
[root@server conf.d]# firewall-cmd --add-service=https --permanent
success
[root@server conf.d]# firewall-cmd --reload
success
[root@server conf.d]# systemctl restart httpd
Job for httpd.service failed because the control process exited with error code.
See "systemctl status httpd.service" and "journalctl -xeu httpd.service" for det
ails.
[root@server conf.d]#
```

Конфигурирование PHP

Создание index.php



Проверка phpinfo()

```
root@server:/etc/httpd/conf.d

[root@server conf.d]# systemctl restart httpd
Job for httpd.service failed because the control process exited with error code.
See "systemctl status httpd.service" and "journalctl -xeu httpd.service" for details.
[root@server conf.d]# dnf -y install php
Last metadata expiration check: 0:56:54 ago on Thu 05 Mar 2026 06:39:58 PM UTC.
Dependencies resolved.
=====
Package                Arch      Version              Repository           Size
=====
Installing:
php                    x86_64    8.0.30-5.el9_7      appstream            8.8 k
Installing dependencies:
nginx-filesystem      noarch    2:1.20.1-24.el9     appstream            9.2 k
php-common            x86_64    8.0.30-5.el9_7      appstream            667 k
Installing weak dependencies:
php-cli               x86_64    8.0.30-5.el9_7      appstream            3.1 M
php-fpm               x86_64    8.0.30-5.el9_7      appstream            1.6 M
php-mbstring          x86_64    8.0.30-5.el9_7      appstream            468 k
php-opcache            x86_64    8.0.30-5.el9_7      appstream            510 k
php-pdo                x86_64    8.0.30-5.el9_7      appstream            82 k
php-xml                x86_64    8.0.30-5.el9_7      appstream            133 k

Transaction Summary
=====
Install 9 Packages

Total download size: 6.5 M
Installed size: 35 M
Downloading Packages:
(1/9): nginx-filesystem-1.20.1-24.el9.noarch.rpm 29 kB/s | 9.2 kB 00:00
(2/9): php-8.0.30-5.el9_7.x86_64.rpm            19 kB/s | 8.8 kB 00:00
(3/9): php-cli-8.0.30-5.el9_7.x86_64.rpm         3.1 MB/s | 3.1 MB 00:00
(4/9): php-common-8.0.30-5.el9_7.x86_64.rpm      536 kB/s | 667 kB 00:01
(5/9): php-mbstring-8.0.30-5.el9_7.x86_64.rpm    536 kB/s | 468 kB 00:00
(6/9): php-opcache-8.0.30-5.el9_7.x86_64.rpm     860 kB/s | 510 kB 00:00
(7/9): php-fpm-8.0.30-5.el9_7.x86_64.rpm         931 kB/s | 1.6 MB 00:01
(8/9): php-pdo-8.0.30-5.el9_7.x86_64.rpm        245 kB/s | 82 kB 00:00
(9/9): php-xml-8.0.30-5.el9_7.x86_64.rpm        793 kB/s | 133 kB 00:00
```

Выводы по проделанной работе

Был настроен веб-сервер Apache для работы по протоколу HTTPS с использованием самоподписанного сертификата.
Добавлена поддержка PHP и выполнена проверка его работы.
Обновлены файлы внутреннего окружения Vagrant для сохранения всех выполненных изменений.